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BACHELOR THESIS

Numerical Methods for Short-Term Interest Rate Modeling and
Applications to Pricing Swaps on LIBOR

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Introduction

Accurate models for short-term interest rates are essential for both market participants and financial institutions, such as hedge funds, commercial banks and others. Short-term interest rates serve as a benchmark for various financial instruments, such as floating-rate loans, bonds, derivatives, and interest rate futures. A precise understanding of the behavior and dynamics of short-term interest rates allows financial institutions to appropriately price these instruments, manage risk, and make informed investment decisions.

In addition, short-term interest rate modeling is vital for understanding the overall macroeconomic environment. Short-term interest rates directly influence consumption, investment, and borrowing costs, which, in turn, impact economic growth and inflation. Considering all the above, using advanced mathematical tools, such as SDE and numerical probability, could be helpful in building consistent predictive models and, hence, help financial institutions and market participants make solid judgements.

The modeling of short-term interest rates is a critical area of research that poses several challenges in proper model selection. Despite the extensive body of literature on models, parameters estimation techniques and calibration methods, that are suitable for interest rate modeling, the main objective is to find the best combination of all the above in a sense of minimizing possible error between the real data and chosen model.

The interest rate swaps are one of the most liquid and interest rate derivatives, followed by the fact that its mechanism is simple enough compared to another derivatives. Hence pricing interest rate swaps is important task to solve both for financial application and for achieving theoretical results in financial mathematics.

The main tasks of the research are:

1. To review recent papers regarding short-term interest rate modeling and SDE's parameter estimation
2. To select appropriate models that can be used for the research's objective
3. To test several approximation methods and apply them to get a numerical approximation of a solution to the model picked previously
4. To choose efficient SDE's parameter estimation techniques and test them for each model picked in Point 2
5. To collect day-by-day data on Libor and SOFR interest rates and fit the best model on it
6. To make a forecast of short term interest rate using the fitted model and price swaps on these rates.

1 Model setup

Since the aim of this project is interest-rate modeling, general view of the models and literature review will be discussed in this section. Also we will cover some basic definitions and most common way to simulate a solution of a stochastic differential equation.

1.1 Preliminaries

It is common to model short-term interest rate dynamic with stochastic differential equations (SDEs), which has the following form:

$$\begin{cases} dX_t = f(X_t)dt + g(X_t)dW_t \\ X_0 = x_0 \end{cases} \quad (1)$$

where $f(t, x)$ is a drift function, $g(t, x)$ is a volatility function, W_t is a multi-dimensional (in general) Brownian motion, which corresponds to uncertainty in a short interest rate dynamic. Here are some mathematical definitions that are needed in the work:

Definition 1.1 (Standard Brownian motion (see e.g. Higham, 2001)). The continuous-time stochastic process $(W_t, t \geq 0)$ is said to be a standard $\mathcal{F}_t$ -Brownian motion if the following properties are satisfied:

1. $W_0 = 0$ a.s.
2. $(W_t, t \geq 0)$ has continuous paths a.s.
3. for all $0 \leq s \leq t < \infty$, $W_t - W_s$ is independent to $\mathcal{F}_s$
4. for all $0 \leq s < t < \infty$,

$$W_t - W_s = W_{t-s} \sim \mathcal{N}(0, t - s)$$

Definition 1.2 (local Lipschitz continuity). Function f satisfies the local Lipschitz condition if for any $R > 0$, there exists $K_R > 0$ such that

$$|f(x) - f(y)| \leq K_R |x - y|$$

for all $x, y \in \mathbb{R}$ with $\max(|x|, |y|) \leq R$.

Definition 1.3 (one-sided Lipschitz continuity). Function f satisfies the one-sided Lipschitz condition if there exists a constant $L > 0$ such that

$$(x - y)(f(x) - f(y)) \leq L|x - y|^2$$

for all $x, y \in \mathbb{R}$.

Definition 1.4 (Hölder continuity). Function f satisfies the Hölder continuity condition if there exist constants $L > 0$ and $0 \leq \alpha < \frac{1}{2}$ such that

$$|f(x) - f(y)| \leq L|x - y|^{\frac{1}{2} + \alpha}$$

for all $x, y \in \mathbb{R}_+$.

Definition 1.5 (Lipschitz continuity). Function f is globally Lipschitz continuous if there exists constant $C \in \mathbb{R}_+$ such that

$$|f(x) - f(y)| \leq C|x - y|$$

for all $x, y \in \mathbb{R}$.

Definition 1.6 (γ -locally Lipschitz continuity). Let f be a real-valued function and $dom(f)$ denotes its definition domain. Function f is γ -locally Lipschitz if there exist non-negative constants C and γ such that

$$|f(x) - f(y)| \leq C(1 + |x|^\gamma + |y|^\gamma)|x - y|, \forall x, y \in dom(f).$$

Definition 1.7 (Ito formula). Let $(M_t, t \geq 0)$ be a continuous square integrable $\mathcal{F}_t$ -martingale and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with continuous derivatives up to order 2. Then

$$f(M_t) = f(M_0) + \int_0^t f'(M_s) dM_s + \frac{1}{2} \int_0^t f''(M_s) d\langle M \rangle_s, t \geq 0.$$

1.2 Vasicek model

Short-term interest rates are used to measure the yield of short-term borrowings taking place in the market. By analyzing the behavior of the interest rate curve, we can predict its further changes and get information about the behavior of financial markets. Historically, one of the first frameworks for interest rate modeling is the Vasicek model, presented in (Vasicek, 1977). Author models default-free market of bonds, where spot rate X_t is a one-dimensional stochastic process under two assumptions:

1. X_t has continuous paths a.s.
2. X_t is a Markov process

Under these assumptions, interest rate dynamic is chosen to be given by the SDE:

$$\begin{cases} dX_t = a(b - X_t)dt + \sigma dW_t \\ X_0 = x_0 \end{cases} \quad (2)$$

where $b, \sigma \in \mathbb{R}$, $a, x_0 \in \mathbb{R}_+$. Process (2) is also called an Ornstein-Uhlenbeck process and its solution has a stationary distribution. This model satisfies basic assumptions for short-term interest rate modeling, such as:

- **Mean reversion.** Under the definition (2) parameter b can be interpreted as a mean of the process, to which the drift function $f(x) = a(b - x)$ pushes the process back.
- **Constant volatility.** The volatility of the process is constant since $g(x) = \sigma$ does not depend on the value of the process. Under this framework we can consider stochastic volatility as a fluctuations around the theoretical mean b . This behaviour is not confirmed by the empirical data on short term interest rates, since when its value sharply increases, uncertainty on the market increases comparably. The form of the volatility function will be extended in the models considered in next chapters.

1.3 CIR process

The CIR process, widely used for short interest rate modeling, had been introduced in (Cox et al, 1985). It is defined by the SDE:

$$\begin{cases} dX_t = (a - kX_t)dt + \sigma\sqrt{X_t}dW_t \\ X_0 = x_0 \end{cases} \quad (3)$$

where $a, k, \sigma \in \mathbb{R}$, $x_0 \in \mathbb{R}_+$. If we put k out of the brackets, we can get

$$dX_t = k\left(\frac{a}{k} - X_t\right)dt + \sigma\sqrt{X_t}dW_t \quad (4)$$

where $\frac{a}{k}$ can be interpreted as a mean of the process, k as a speed of reaching the mean and σ as a measure of volatility.

Since CIR is used for interest-rate modeling, it is natural to assume that X_t is always nonnegative. This condition is achieved when $2a > \sigma^2$, in this case time increment always outweigh possible negative impact of volatility part. This condition also ensures that if X_t reaches 0, it will instantly become positive. Starting from here we assume that this condition is always satisfied.

Other practically relevant properties of the CIR process are increasing volatility as X_t itself increases and existence and uniqueness of a steady state distribution. The first property corresponds to the fact that derivative of the volatility function is always positive.

1.4 Ait-Sahalia model

Ait-Sahalia model (Ait-Sahalia, 1996) is given by the class of SDEs of the form:

$$\begin{cases} dX_t = (\alpha_0 + \alpha_1 X_t + \alpha_2 X_t^2 + \frac{\alpha_3}{X_t})dt + \sqrt{\beta_0 + \beta_1 X_t + \beta_2 X_t^{\beta_3}}dW_t \\ X_0 = x_0 \end{cases} \quad (5)$$

where SDE depends on the of parameters $\theta = (\alpha_0, \alpha_1, \alpha_2, \alpha_3, \beta_0, \beta_1, \beta_2, \beta_3) \in \mathbb{R}^8$, $x_0 \in \mathbb{R}_+$. As for the CIR model, it is natural to imply restrictions on the coefficients in order to $X_t \geq 0$ for all $t \in \mathbb{R}_+$. These conditions are given as follows:

$$\beta_0 \geq 0 \text{ with (if } \beta_0 = 0, (\beta_2 > 0, \beta_3 \in (0, 1) \text{ or } (\beta_1 > 0, \beta_3 > 1)) \quad a$$

and

$$\beta_2 > 0 \text{ if either } \beta_3 > 1 \text{ or } \beta_1 = 0 \quad b$$

and

$$\beta_1 > 0 \text{ if either } 0 < \beta_3 < 1 \text{ or } \beta_2 = 0 \quad c$$

and

$$\alpha_2 \leq 0 \text{ with } \alpha_1 < 0 \text{ if } \alpha_2 = 0 \quad d$$

and

$$(\alpha_3 > 0, 2\alpha_3 \geq \beta_0 \geq 0) \text{ or } (\alpha_3 = 0, \alpha_0 > 0, \beta_0 = 0, \beta_3 > 1 \text{ and } 2\alpha_0 \geq \beta_1 > 0) \quad e$$

This model can be naturally considered as an extension of a large class of models that are used to short interest rate modeling. Some of these models and corresponding drift and volatility functions are represented in the table below:

Model	Drift function $f(x, \theta)$	Diffusion function $g^2(x, \theta)$
Vasicek model (Vasicek, 1977)	$\alpha_0 + \alpha_1 x$	β_0
CIR model (Cox et al, 1985)	$\alpha_0 + \alpha_1 x$	$\beta_1 x$

Table 1: Special cases of Ait-Sahalia model. Source: (Ait-Sahalia, 1996)

Since Ait-Sahalia model is a large class of SDEs with a lot of parameters, we have to specify particular SDE to consider:

$$\begin{cases} dX_t = bX_t dt + \sigma X_t^\beta dW_t \\ X_0 = x_0 \end{cases} \quad (6)$$

where $\alpha_0 = \alpha_2 = \alpha_3 = \beta_0 = \beta_1 = 0$, $b = \alpha_1$, $\sigma = \sqrt{\beta_2}$, $\beta = 2\beta_3$. Model with these coefficients satisfies conditions $b - e$. In (Ait-Sahalia, 1996) it is written that condition a is needed in order to diffusion function of the process X_t stays positive for any x . Since $g(x, \theta) = \sigma x^\beta$ and x is always positive, we can take $\beta_1 = 0$. Starting from here, by Ait-Sahalia model we mean (6). Choosing this specific form of the model is caused by several reasons, such as:

- β , power of the volatility function, is a parameter that can be estimated. Hence volatility can be predicted better compared to CIR process, where power of the volatility function is set to $\frac{1}{2}$
- Drift function $f(x) = b \cdot x$ is equal to 0 at 0, it is useful when applying Exponential scheme to the model.

1.5 Euler-Maruyama scheme

The simplest way to simulate a stochastic differential equation is to use Euler-Maruyama method, that comes from Euler method for the simulation of deterministic ordinary differential equations. We will introduce the scheme according to (Higham, 2001) and consider its application to CIR process simulation. In order to explain how the scheme works, first of all let's write down the form of stochastic differential equation that we will simulate:

$$dX_t = f(X_t)dt + g(X_t)dW_t, X_0 = x_0, 0 \leq t \leq T, \quad (7)$$

where $x_0 \in \mathbb{R}$, f and g are scalar functions and T is a finite-time horizon of observations.

First of all we should choose a positive integer N and split the interval $[0; T]$ into N equal parts, so each part will be the size of $\Delta := \frac{T}{N}$. Taking $t_i = i\Delta$, now for the discrete timesteps we can write from (7), that:

$$X_{t_i} = X_{t_{i-1}} + \int_{t_{i-1}}^{t_i} f(X_s)ds + \int_{t_{i-1}}^{t_i} g(X_s)dW_s \quad (8)$$

For $t_i - t_{i-1}$ small enough we may consider approximation

$$\int_{t_{i-1}}^{t_i} g(X_s)dW_s \sim g(X_{t_{i-1}})(W_{t_i} - W_{t_{i-1}}), \quad (9)$$

so finally we get the sequence of random variables $\hat{X}_0, \hat{X}_1, \hat{X}_2, \dots, \hat{X}_L$, where:

$$\hat{X}_0 = x_0, \hat{X}_i = \hat{X}_{i-1} + f(\hat{X}_{i-1})\Delta + g(\hat{X}_{i-1})(W_{t_i} - W_{t_{i-1}}) \quad (10)$$

We will call this sequence the discrete EM approximation for the SDE (7).

After introducing this scheme we can test it empirically: let's take the geometric brownian motion:

$$\begin{cases} dX_t = \mu X_t dt + \sigma X_t dW_t \\ X_0 = x_0 \end{cases} \quad (11)$$

for which we know the exact solution:

$$X_t = x_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right) \quad (12)$$

and simulate it with EM method.

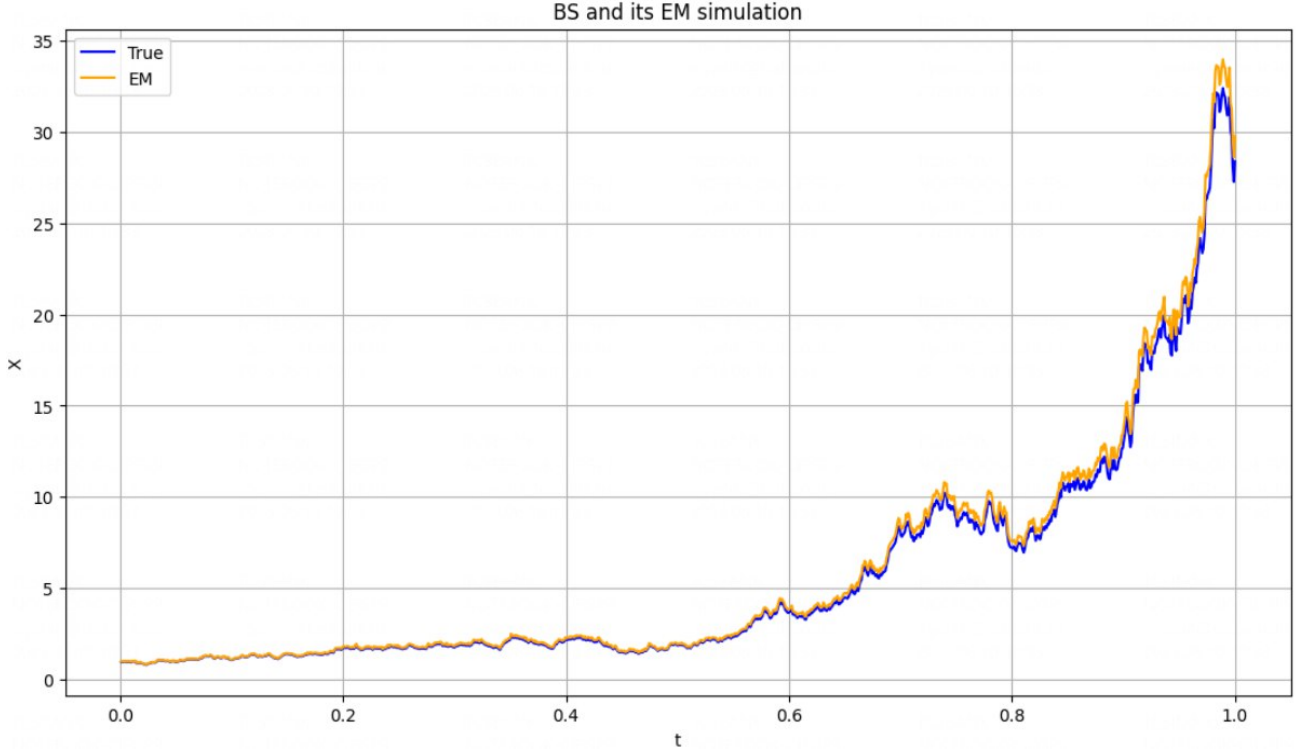


Figure 1: Black-Scholes model and EM simulation. Source: author's calculations

Now we can visually check, that EM simulation coincides with the true solution of an SDE, but we have to specify strict criteria of similarity between two stochastic processes. Further in the work we will use strong convergence to check if the initial process and simulation converge to each other and if so, at what speed.

Definition 1.8 (Strong convergence). A numerical scheme $\hat{X}$ strongly converges to the process X if there exist constants p , C and γ such that

$$\sup_{n \in [0, N]} \sup_{t \in [n\Delta, (n+1)\Delta]} \mathbb{E}[|X_t - \hat{X}_n|^p] \leq C(\Delta)^{\gamma p} \quad (13)$$

We will call constant γ a rate of convergence.

The intuition behind this definition is quite simple: we should calculate expected difference between initial process and its simulation at all points where our scheme exists and take the maximum error. The expected value of $|X_t - \hat{X}_n|$ will depend on the timestep Δ . We must try to bound this function with monomial of degree γ : the greater γ the faster numerical scheme converge to the true solution. If we can not bound resulting function with monomial (for example $\mathbb{E}|X_t - \hat{X}_n| \propto e^\Delta$), we will say that our scheme diverges.

Strong convergence is not the only way to measure difference between two stochastic process. There are tasks where we are interested not in difference between two processes itself but in difference of the function of the processes, where weak convergence is needed.

Definition 1.9 (Weak convergence). A numerical scheme $\hat{X}$ weakly converges to the process X if there exist constants p , C and γ such that for any smooth function h with polynomial growth condition,

$$\sup_{n \in [0, N]} \sup_{t \in [n\Delta, (n+1)\Delta]} \mathbb{E}[|h(X_t) - h(\hat{X}_n)|^p] \leq C(\Delta t)^\gamma \quad (14)$$

Example of a task where weak convergence is needed is pricing american call option's value on stock on a finite time horizon $[0, T]$. Define stock's value at time t as S_t , strike price as K , so the payoff at time T will be

$$\sup_{t \in [0, T]} \max(0, S_t - K). \quad (15)$$

Define $p(x) = \sup_{t \in [0, T]} \max(0, x - K)$. If we have numerical approximation $\hat{S}_t$ to S_t , we are interested not in difference between S_t and $\hat{S}_t$, but in difference between $p(\hat{S}_t)$ and $p(S_t)$.

It can be shown that if f and g in (7) are Lipschitz continuous, the rate of strong convergence of Euler-Maruyama scheme is equal to $\frac{1}{2}$. But what if we try to use this scheme for CIR process? Let's put the coefficients into (10), so we get the scheme:

$$\hat{X}_i = \hat{X}_{i-1} + (a - b\hat{X}_{i-1})\Delta t + \sigma\sqrt{\hat{X}_{i-1}}(W_{t_i} - W_{t_{i-1}}) \quad (16)$$

EM scheme does not imply any restrictions on the solution, so in general it can be negative or even infinite depending on properties of simulated process. Since volatility function of the CIR process is a square root, its approximated solution must be nonnegative on the whole interval $[0, T]$. Hence we should extend the basic EM method to control the solution of the scheme.

2 Advanced numerical schemes

Strong convergence for the Euler-Maruyama scheme requires Lipschitzness of f and g . One of the core models of this work, the CIR process, has a volatility function given by $g(x) = \sigma\sqrt{x}$ that is not Lipschitz. It means that we have to consider other numerical schemes which can be applied to a larger class of SDEs.

2.1 Truncated Euler scheme

A first modification of basic Euler-Maruyama method we are about to discuss is presented in (Yang et al, 2020). The idea of this method is to "truncate" the initial drift and volatility functions, f and g , so we can control the moments of the process. First of all we should make some assumptions on the coefficients and make sure that they are satisfied for a CIR process (3).

Assumptions to be considered are:

- Drift function f is locally Lipschitz continuous (Definition 1.2), satisfies one-sided Lipschitz condition (Definition 1.3) and there exists $\gamma > 0$ and $H > 0$ such that

$$|f(x) - f(y)|^2 \leq H(1 + |x|^\gamma + |y|^\gamma)|x - y|^2$$

- Diffusion function g is Hölder continuous (Definition 1.4).

Statement 2.1. Function $f(x) = a - kx$ satisfies the local Lipschitz condition.

Proof. $|f(x) - f(y)| = |a - kx - a + ky| = |k| \cdot |x - y| \Rightarrow$ local Lipschitz condition is satisfied with $K_R = |k|$. $\square$

Statement 2.2. Function $f(x) = a - kx$ satisfies the one-sided Lipschitz condition.

Proof. $(x - y)(f(x) - f(y)) = (x - y) \cdot (a - kx - a + ky) = k \cdot |x - y|^2 \Rightarrow$ one-sided Lipschitz condition is satisfied with $L = |k|$. $\square$

Statement 2.3. For function $f(x) = a - kx$ there exists $\gamma > 0$ and $H > 0$ such that

$$|f(x) - f(y)|^2 \leq H(1 + |x|^\gamma + |y|^\gamma)|x - y|^2$$

for all $x, y \in \mathbb{R}$.

Proof.

$$(f(x) - f(y))^2 = k^2 \cdot (x - y)^2 \leq H(1 + |x|^\gamma + |y|^\gamma)|x - y|^2$$

$\forall \gamma > 0$ and $H = k^2$. $\square$

Statement 2.4. Function $g(x) = \sigma\sqrt{x}$ satisfies the Hölder continuity condition.

Proof.

$$|g(x) - g(y)| = \sigma|\sqrt{x} - \sqrt{y}| = \sigma \frac{|x - y|}{\sqrt{x} + \sqrt{y}} = \sigma|x - y|^{\frac{1}{2}} \frac{\sqrt{|x - y|}}{\sqrt{x} + \sqrt{y}}$$

$$(\sqrt{|x - y|})^2 = |x - y| \leq |x| + |y| = (\sqrt{|x|} + \sqrt{|y|})^2 \leq (\sqrt{x} + \sqrt{y})^2 \Rightarrow \sqrt{|x - y|} \leq \sqrt{x} + \sqrt{y}$$

$$|g(x) - g(y)| = \sigma|x - y|^{\frac{1}{2}} \frac{\sqrt{|x - y|}}{\sqrt{x} + \sqrt{y}} \leq \sigma|x - y|^{\frac{1}{2}}$$

Holder continuity condition is satisfied with $L = \sigma$ and $\alpha = 0$ □

In (Yang et al, 2020) it is proved that for any process $dx(t) = f(x(t))dt + g(x(t))dW(t)$ whose drift coefficient are locally Lipschitz continuous and satisfy one-sided Lipschitz condition:

$$\mathbb{E}[\sup_{0 \leq t \leq T} |x(t)|^p] < \infty \text{ for any } p > 0.$$

Now we can start truncating initial functions to get the scheme. First we have to choose function $\mu : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that

$$\sup_{|x| < u} |f(x)| \leq \mu(u), \forall u \geq 1.$$

Since our scheme is defined in $\mathbb{R}_+$, we may take $\mu(u) = a \cdot u + a$. Now we can denote μ^{-1} by the inverse function of μ . This function is strictly increasing and equal to $\frac{x-a}{a}$ at the point x . Then we choose parameter $\Delta = \frac{N}{L}$. Here and further we will take $T = 1$ and $L = 2^{10}$, hence $\Delta t = \frac{T}{L} = \frac{1}{2^{10}}$. We also should define function $h : (0, \Delta] \rightarrow (0, \infty)$ such that

$$h(\Delta) \geq \mu(2), \lim_{\Delta \rightarrow 0} h(\Delta) = \infty \text{ and } \Delta^{\frac{1}{4}} h(\Delta) \leq 1, \forall \Delta \in (0, \Delta^*]$$

Such function can be defined by $h(\Delta) = \frac{3a}{\Delta^{\frac{1}{6}}}$, that satisfies all the condition above. For a given Δ we can define our truncated function f as

$$f_{\Delta}(x) = f(\min(|x|, \mu^{-1}(h(\Delta))) \cdot \text{sign}(x)). \quad (17)$$

Since function $g(x)$ satisfies linear growth condition ($g(x) = \sigma\sqrt{x} \leq \sigma x + 1, \forall x \in \mathbb{R}_+$) there is no need to truncate it. Having $f_{\Delta}(x) = a - k \cdot \min(x, \frac{3}{\Delta^{\frac{1}{6}}} - 1)$, the truncated scheme for the CIR process can be defined as

$$\hat{X}_{i+1} = \hat{X}_i + f_{\Delta}(\hat{X}_i)\Delta + g(\hat{X}_i)(W_{t_{i+1}} - W_{t_i}), \hat{X}_0 = x_0 \quad (18)$$

We can use it to simulate CIR process with parameters $a = 2$, $k = 1$, $\sigma = 1$ and $x_0 = 1$:

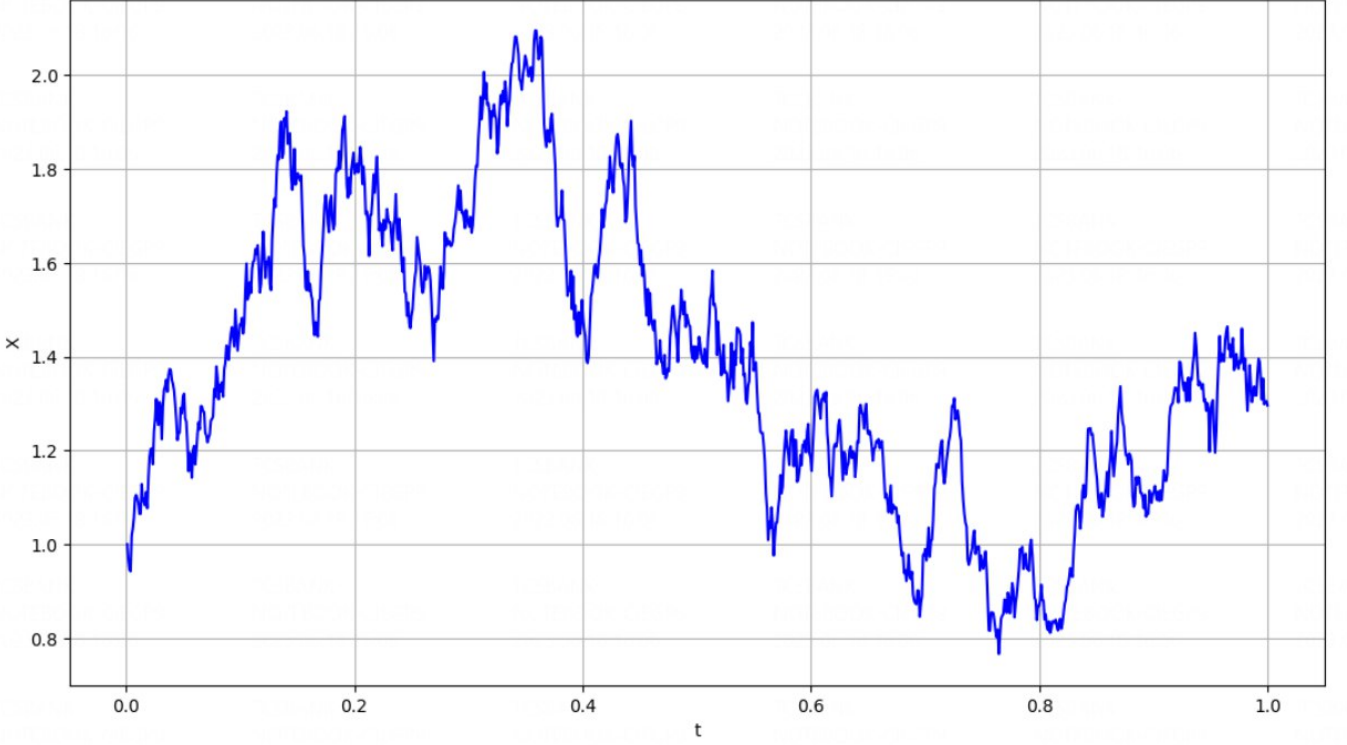


Figure 2: CIR process simulation with Truncated EM. Source: author's calculations

We can visually check that the scheme converge to its theoretical mean $\frac{a}{k} = 2$. Also the trajectory is volatile enough because $\sigma = 1$. Hence we can empirically check that our scheme works, but we should prove it formally and find the rate of convergence. In (Yang et al, 2020) it is shown that for process with coefficients f and g , that satisfies all the assumptions above,

$$\mathbb{E}[\sup_{0 \leq t \leq T} |X_t - \hat{X}_t|] \leq C(\frac{1}{\ln \Delta^{-1}})^{\frac{1}{2}} \quad (19)$$

Note that generally speaking $\mathbb{E}[\sup \dots] \neq \sup \mathbb{E}[\dots]$, that's why we will use these two types of strong convergence and compare them correctly.

2.2 Tamed Euler scheme

Let us now discuss the next scheme, that is called Tamed Euler scheme. It was first proposed in (Hutzenthaler et al, 2011). Before explaining the idea of the scheme we should check if all initial conditions from the paper are satisfied: drift coefficient should be one-sided Lipschitz continuous (Definition 1.3) and its derivative must have at least polynomial growth. Diffusion coefficient should be globally Lipschitz continuous (Definition 1.5). The fact that $f(x) = a - kx$ is one-sided Lipschitz continuous is checked in the previous section. Since drift coefficient is linear function, it's derivative is a constant, so it grows less then polynomially.

Statement 2.5. Drift function $g(x) = \sigma\sqrt{x}$ is not globally Lipschitz continuous

Proof. Let's take $y = 0$ and check if there exists a constant such that $\sqrt{x} \leq C|x|$ for all $x \in \mathbb{R}_+$. On

interval $(0, 1)$ this inequality is equivalent to $1 \leq C\sqrt{x}$. On this interval we can take any $C \in \mathbb{R}_+$ and see that there always exists x from $\mathbb{R}_+$ such that $1 > C\sqrt{x}$ because $\lim_{x \rightarrow 0} \sqrt{x} = 0$.

□

Two out of three conditions are not satisfied, so formally we can not use this scheme for CIR process modeling. But nevertheless we will describe the method in order to apply it to another SDE in the future and even try to use with CIR process to check what will go wrong.

The idea of method is as follows: we will bound our drift function f , but we will do it differently from the previous section, taking:

$$f_{\Delta}(x) = \frac{f(x)}{1 + \Delta \cdot |f(x)|} \quad (20)$$

It is easy to see that $|f_{\Delta}(x)| < 1, \forall x \in \mathbb{R}$, so we can prevent our scheme from diverging since drift function is tamed and diffusion function grows less than linearly. Now we can write down the discrete scheme:

$$\hat{X}_{i+1} = \hat{X}_i + \frac{\Delta \cdot f(\hat{X}_i)}{1 + \Delta \cdot |f(\hat{X}_i)|} + g(\hat{X}_i)(W_{t_{i+1}} - W_{t_i}) \quad (21)$$

Since the scheme is ready we try to use it for CIR process modeling with $a = 2$, $k = 1$ and $\sigma = 1$ (recall that formally we are not allowed to do it since initial conditions are not satisfied):

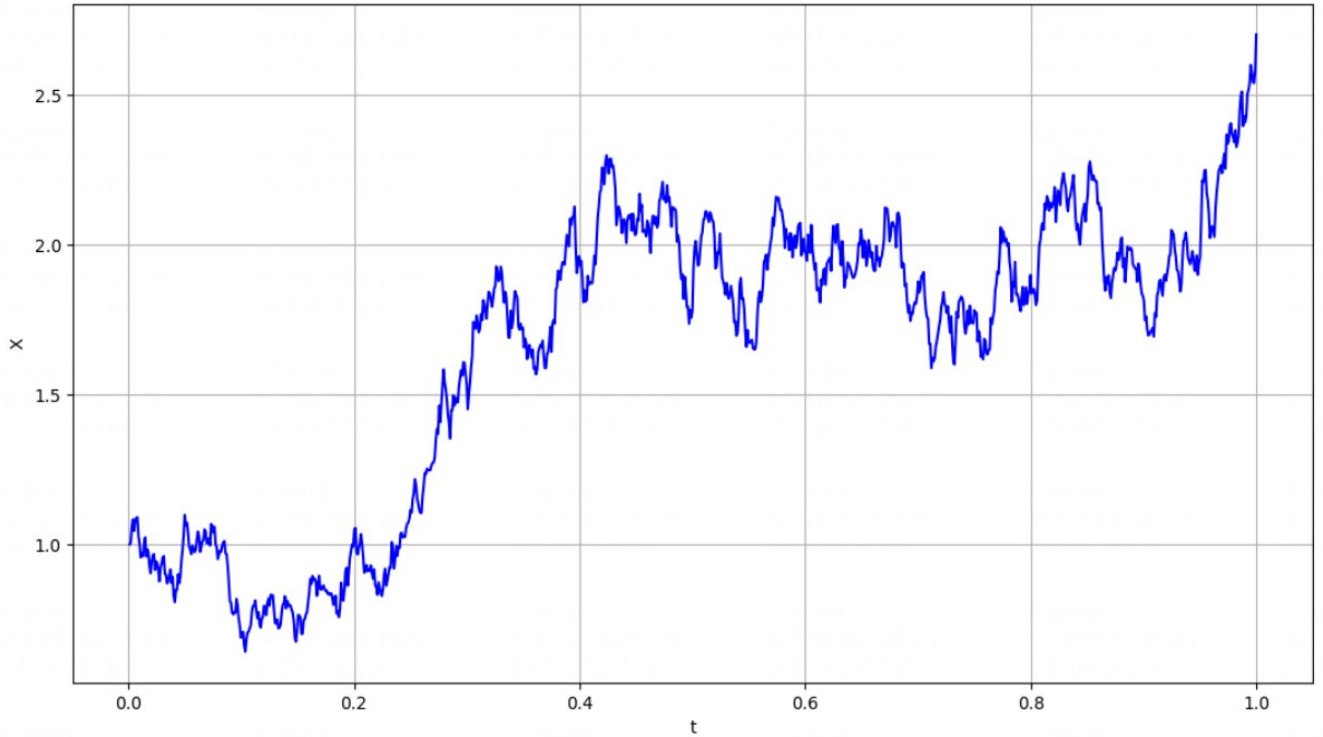


Figure 3: CIR process simulation with Tamed EM. Source: author's calculations

As we can see, graph did not approach 0, so we may say that the scheme worked out for the CIR process with parameters written above. Our hypothesis is that the Tamed scheme converges fast enough, so taking $N = 2^{10}$ allows to use coefficients that don't satisfy theoretical conditions, the scheme anyway will converge to initial process that is nonnegative by definition. We can visually test it by simulating same process with different number of timesteps (Figure 4).

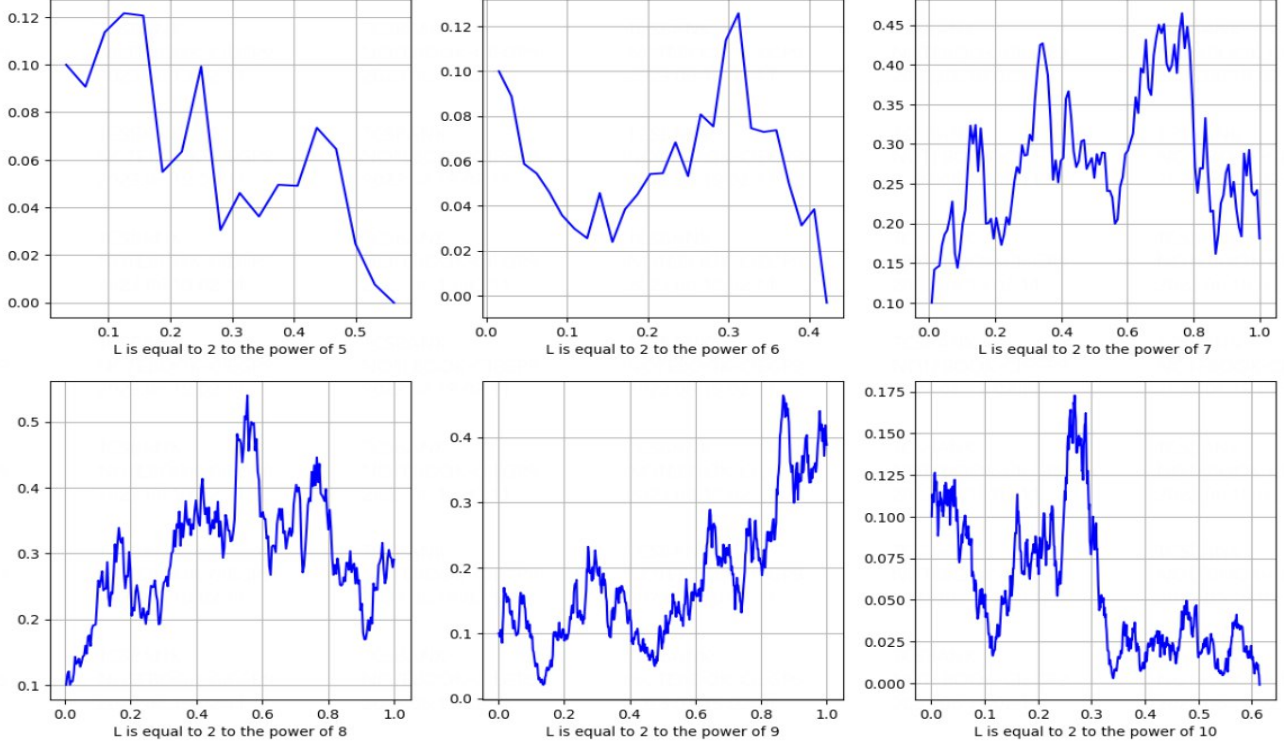


Figure 4: CIR process simulation with Truncated EM and different number of timesteps. Source: author's calculations

As we can see schemes with number of timesteps equal to 2 to the power of 5, 6 and 10 are failed to converge. We may suppose that Tamed scheme will work with CIR process if x_0 and a are far enough from 0, and σ is not too large compared to a and k .

In order to measure the strong convergence we may refer to (Hutzenthaler et al, 2011):

$$\mathbb{E} \left[\sup_{t \in [0, T]} |X_t - \bar{X}_t| \right] \leq C \cdot L^{-\frac{1}{2}} \quad (22)$$

2.3 Implicit scheme by Alfonsi [1]

The last scheme we are going to use is derived in (Alfonsi, 2005). The author developed his scheme only for the CIR process (3). The idea of method is to try to expand the implicit scheme into quadratic equation and choose positive solution:

$$X_t = x_0 + \int_0^t (a - kX_s) ds + \sigma \int_0^t \sqrt{X_s} dW_s$$

$$\begin{aligned}
&= x_0 + \lim_{n \rightarrow \infty} \left(\sum_{t_i < t} (a - kX_{t_{i+1}}) \Delta + \sigma \sum_{t_i < t} \sqrt{X_{t_{i+1}}} (W_{t_{i+1}} - W_{t_i}) - \sigma \sum_{t_i < t} (\sqrt{X_{t_{i+1}}} - \sqrt{X_{t_i}}) (W_{t_{i+1}} - W_{t_i}) \right) \\
&= x_0 + \lim_{n \rightarrow \infty} \left(\sum_{t_i < t} (a - \frac{\sigma^2}{2} - kX_{t_{i+1}}) \Delta + \sigma \sum_{t_i < t} \sqrt{X_{t_{i+1}}} (W_{t_{i+1}} - W_{t_i}) \right) \tag{23}
\end{aligned}$$

Based on this expression we can write down a quadratic equation, which positive solution will be our discrete scheme:

$$\hat{X}_{i+1} = X_i + (a - \frac{\sigma^2}{2} - k\hat{X}_{i+1})\Delta + \sigma\sqrt{\hat{X}_{i+1}}(W_{t_{i+1}} - W_{t_i}) \tag{24}$$

Based on our previous calculations we can say that $\hat{X}_{t_i} \rightarrow X_{t_i}$ in almost sure sense, as number of timesteps N goes to infinity. So we may use it with $\Delta = \frac{T}{N}$ small enough. Hence we choose positive solution from (24):

$$\hat{X}_{i+1} = \left(\frac{\sigma(W_{t_{i+1}} - W_{t_i}) + \sqrt{\sigma^2(W_{t_{i+1}} - W_{t_i})^2 + 4(\hat{X}_i + (a - \frac{\sigma^2}{2})\Delta)(1 + k\Delta)}}{2(1 + k\Delta)} \right)^2 \tag{25}$$

Formally we got a scheme, that is obviously nonnegative, but it's hard to calculate. Later we will try to expand it, using Taylor series and some properties, that have been proven in the paper.

Statement 2.6. Suppose that $(\hat{X}_i)$ is a nonnegative scheme, for which

$$\begin{cases} \hat{X}_0 = x_0 \\ \hat{X}_{i+1} \leq (1 + \frac{k}{N})\hat{X}_i + \sigma\sqrt{\hat{X}_i}(W_{t_{i+1}} - W_{t_i}) + \mathcal{O}(\frac{1}{L}) \end{cases} \tag{26}$$

is satisfied $\forall i \leq N - 1$. Then, scheme $(\hat{X}_i)$ has uniformly bounded moments, so it means that $\hat{X}_i = \mathcal{O}(1) \forall i \in [1, N]$.

The proof of this statement is complicated enough to use this fact only referring to (Alfonsi, 2005). We will use it to reduce some parts of the scheme to $\mathcal{O}(1)$. Now we can define hypothesis H_1 to explore strong convergence:

We will say that scheme $(\hat{X}_i)$ satisfies H_1 if it is nonnegative scheme such that:

$$\hat{X}_{i+1} = \hat{X}_i + (a - k\hat{X}_i)\Delta + \sigma\sqrt{\hat{X}_i}(W_{t_{i+1}} - W_{t_i}) + m_{t_1} - m_i + \mathcal{O}(\frac{1}{N^{\frac{3}{2}}}) \tag{27}$$

where $m_{t_{i+1}} - m_{t_i}$ is a martingale of order 1 ($m_{t_{i+1}} - m_{t_i} = \mathcal{O}(\frac{1}{L})$)

Now we can rewrite our scheme as

$$\hat{X}_{i+1} = \left(\frac{\sigma(W_{t_{i+1}} - W_{t_i}) + \sqrt{\sigma^2(W_{t_{i+1}} - W_{t_i})^2 + 4(\hat{X}_i + (a - \frac{\sigma^2}{2})\Delta)(1 + k\Delta)}}{2(1 + k\Delta)} \right)^2$$

$$\begin{aligned}
&= \frac{1}{4(1+k\Delta)^2} \left(\sigma^2(W_{t_{i+1}} - W_{t_i})^2 + \sigma^2(W_{t_{i+1}} - W_{t_i})^2 + 4(\hat{X}_i + (a - \frac{\sigma^2}{2})\Delta)(1+k\Delta) + \right. \\
&\quad \left. + 2\sigma(W_{t_{i+1}} - W_{t_i})\sqrt{\sigma^2(W_{t_{i+1}} - W_{t_i})^2 + 4(\hat{X}_i + (a - \frac{\sigma^2}{2})\Delta)(1+k\Delta)} \right) \quad (28)
\end{aligned}$$

We may claim that under Statement 2.6

$$\sqrt{\sigma^2(W_{t_{i+1}} - W_{t_i})^2 + 4(\hat{X}_i + (a - \frac{\sigma^2}{2})\Delta)(1+k\Delta)} = \mathcal{O}(\frac{1}{\sqrt{N}}), \quad (29)$$

so the scheme looks like

$$\hat{X}_{i+1} = \frac{1}{1+k\Delta} \hat{X}_i + \frac{\sigma}{(1+k\Delta)^{\frac{3}{2}}} \sqrt{\hat{X}_i} (W_{t_{i+1}} - W_{t_i}) + \mathcal{O}(\frac{1}{N}). \quad (30)$$

Using this fact we can prove that $\hat{X}$ satisfies H_1 as it is done in the paper of Alfonsi. After doing this, we may state, that for the implicit scheme $\hat{X}$:

$$\sup_{0 \leq i \leq n} \mathbb{E}(|X_i - \hat{X}_{t_i}|) \leq \frac{C}{\ln(L)} \quad (31)$$

$$\mathbb{E}[\sup_{0 \leq i \leq n} |X_i - \hat{X}_{t_i}|] \leq \frac{C}{\sqrt{\ln(L)}} \quad (32)$$

Now we can test the scheme for CIR process with $a = 3$, $k = 2$ and $\sigma = 1$:

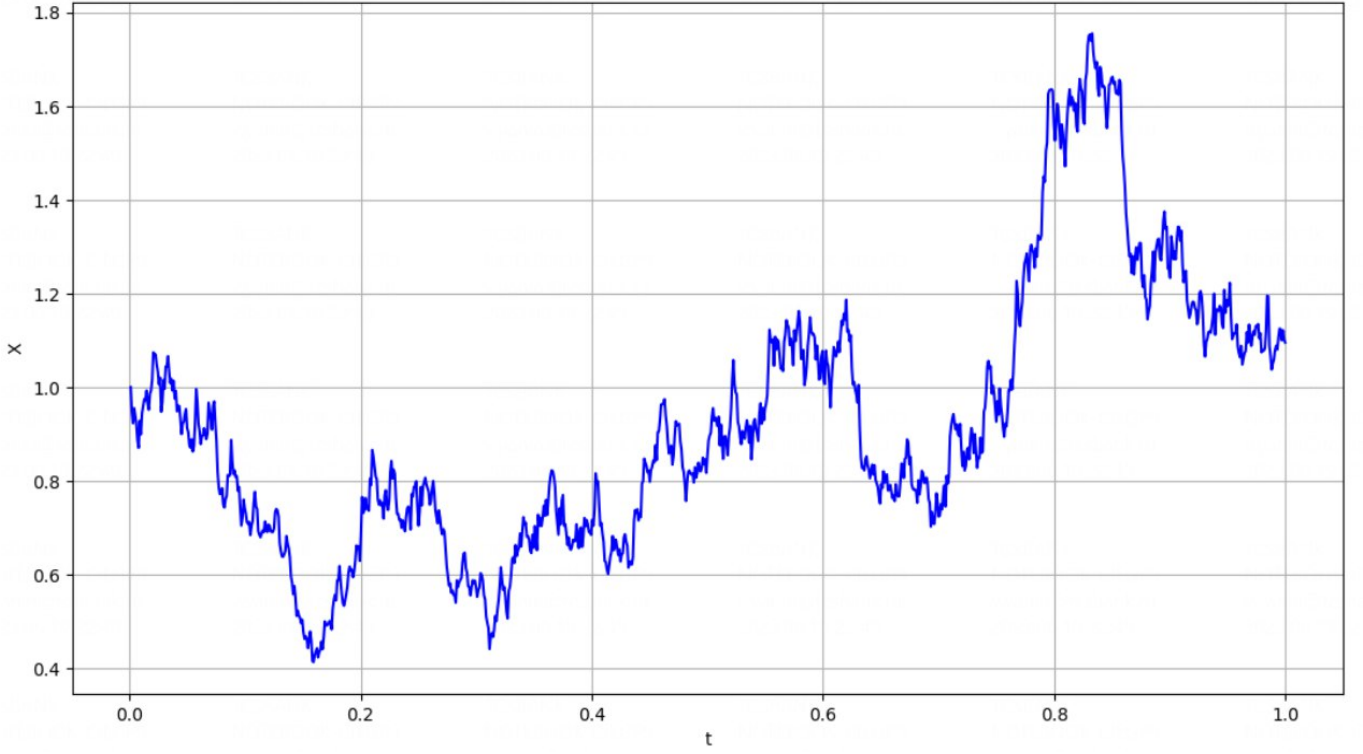


Figure 5: CIR process simulation with Implicit scheme by Alfonsi. Source: author's calculations

2.4 Exponential Euler scheme

Exponential Euler scheme is another advanced numerical scheme proposed in (Bossy et al, 2021). This scheme is designed to approximate solution of one-dimensional SDE with non-globally Lipschitz drift coefficient and monomial diffusion coefficient of order $\beta > 1$. Since an order of the diffusion coefficient of the CIR process is equal to $\frac{1}{2}$, this scheme can be used only for the Ait-Sahalia model:

$$\begin{cases} dX_t = bX_t dt + \sigma X_t^\beta dW_t \\ X_0 = x_0 \end{cases} \quad (33)$$

where $\beta > 1$. Assumptions from (Bossy et al, 2021) should be checked for model (33) in order to be able to use the exponential scheme for this model.

Statement 2.7. Function $f(x) = b \cdot x$ is $2(\beta - 1)$ -locally Lipschitz (Definition 1.7) and $b(0) \geq 0$.

Proof. $|f(x) - f(y)| = |b \cdot x - b \cdot y| = |b| \cdot |x - y| \leq |b| \cdot |x - y| \cdot (1 + |x|^\gamma + |y|^\gamma)$, $b(0) = 0$. □

Statement 2.8. There exist some finite constants $B_1, B_2 \geq 0$, such that, for all $x \geq 0$,

$$f(x) \leq B_1 x - B_2 x^{2\beta-1} + f(0).$$

Proof. $f(x) = b \cdot x = f(0) + b \cdot x \Rightarrow B_1 = b, B_2 = 0$. □

In (Bossy et al, 2021) it is proved that if $\beta > 1$, $\sigma, x_0 > 0$, then with Statements 2.7, 2.8 there exist a unique (strictly) positive strong solution to the SDE (33). Moreover, where exists constant C such that

$$\sup_{t \in [0, T]} \mathbb{E}[X_t^2] \leq C(1 + x_0^2). \quad (34)$$

First let us rewrite (33) as:

$$\begin{cases} dX_t = X_t(bdt + \sigma X_t^{\beta-1} dW_t) \\ X_0 = x_0. \end{cases} \quad (35)$$

Taking $\Delta = \frac{1}{N}$, we consider discrete approximation scheme $(\hat{X}_n, n \geq 1)$ given by:

$$\begin{cases} \hat{X}_{n+1} = \hat{X}_n \cdot \exp\left((b - \frac{\sigma^2}{2} \hat{X}_n^{2(\beta-1)})\Delta + \sigma \hat{X}_n^{\beta-1} (W_{t_{n+1}} - W_{t_n})\right) \\ \hat{X}_0 = x_0. \end{cases} \quad (36)$$

Approximation scheme above can be directly applied to the Ait-Sahalia model (33). Let us take $b = 3, \sigma = 1, \beta = 2$ and $x_0 = 2$. Results of simulation are presented in Figure 6. We can see from the graph that trajectory is more volatile compared to CIR process approximation. It comes from the fact that the power of the diffusion function β is greater than 1, so it enables us capturing possible short-term shocks of an interest rate market when the scheme will be applied to Libor market.

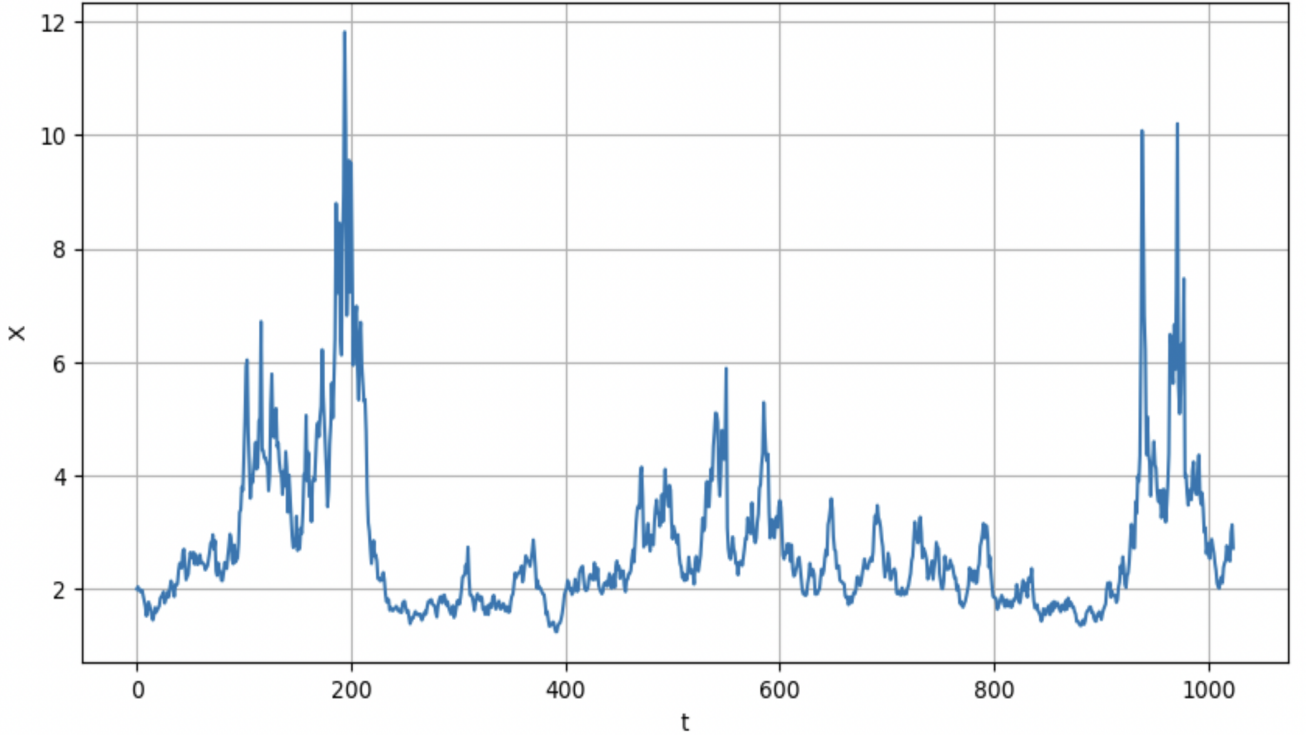


Figure 6: Ait-Sahalia's true solution's approximation with the exponential scheme. Source: author's calculations

In (Bossy et al, 2021) it is stated that under all assumptions above the weak rate of convergence for Ait-Sahalia model (33) with the exponential Euler scheme (36) is equal to one. This rate of convergence is high compared to other numerical schemes. Hence, theoretically, SDE's parameter estimation with exponential scheme approximation of the true solution is more efficient in terms of difference between initial parameters and estimated ones.

3 Parameter estimation techniques

This Section is dedicated to parameter estimation techniques of diffusion processes. Part 1 is based on the samples of the true solution, Part 2 - on the approximation of the true solution with numerical schemes, developed in Section 2.

3.1 Exact Likelihood Inference

Consider the one-dimensional SDE with drift and volatility functions depending on space x and vector of parameters θ :

$$dX_t = \mu(X_t, \theta)dt + \sigma(X_t, \theta)dW_t \quad (37)$$

where $\theta \in \mathbb{R}^n$. Now we define $p_\theta(t, y|x)$ the conditional density of the true solution X_t to (3), given $X_0 = x_0 \in \mathbb{R}_+$. Since μ and σ does not depend on t , $p_\theta(t, y|x)$ is a conditional density of X_{s+t} given $X_s = x$.

Further we assume that values of the initial process are observed in discrete time. For example Libor interest rate is observable in every business day. In this framework we can define time horizon T and split it into N equal intervals $\Delta t = \frac{T}{N}$. Here $t_i = i\Delta$ and $X_i := X_{t_i}$ is a discrete time stochastic process. Now we can define the likelihood function

$$L_N(\theta) = \prod_{i=1}^N p_\theta(\Delta, X_i|X_{i-1})p_\theta(X_0) \quad (38)$$

Hence log-likelihood function $l_N(\theta) = \ln(L_N(\theta))$:

$$l_N(\theta) = \ln(L_N(\theta)) = \sum_{i=1}^N l_i(\theta) + \ln(p_\theta(X_0)) = \sum_{i=1}^N \ln(p_\theta(\Delta, X_i|X_{i-1})) + \ln(p_\theta(X_0)) \quad (39)$$

In order to be able to use Exact Likelihood Estimation, all the assumptions on μ and σ should be satisfied (Lacus, 2008):

1. **Linear growth.** There exists a constant K such that

$$|\mu(x, \theta)| + |\sigma(x, \theta)| \leq K(1 + |x|)$$

for all x .

2. **Global Lipschitzness.** There exists a constant K such that

$$|\mu(x, \theta) - \mu(y, \theta)| + |\sigma(x, \theta) - \sigma(y, \theta)| \leq K|x - y|$$

for all x, y .

3. Positiveness of diffusion coefficient.

$$\inf_x \sigma^2(x, \theta) > 0.$$

4. Bounded moments.

$$\sup_t \mathbb{E}|X_t|^k < \infty$$

for all $k \in \mathbb{N}$.

5. Smoothness of the coefficients.

μ , σ and their derivatives in θ up to order 3 are smooth in x .

Let's consider CIR process of the form:

$$\begin{cases} dX_t = (\theta_1 - \theta_2 X_t)dt + \theta_3 \sqrt{X_t} dW_t \\ X_0 = x_0 \end{cases} \quad (40)$$

where $\theta_1, \theta_2, \theta_3 \in \mathbb{R}_+$ are parameters to be estimated, with condition $2\theta_1 > \theta_3^2$, $x_0 \in \mathbb{R}_+$. We now refer to (Lacus, 2008) in order to write down the conditional density of the true solution:

$$p_\theta(t, y|x) = c \cdot e^{-u-v} \left(\frac{u}{v}\right)^{\frac{q}{2}} I_q(2\sqrt{u \cdot v}) \quad (41)$$

where $c = \frac{2\theta_2}{\theta_3^2(1-e^{-\theta_2 t})}$, $q = \frac{2\theta_1}{\theta_3^2} - 1$, $u = c \cdot x \cdot e^{-\theta_2 t}$, $v = c \cdot y$; $x, y \in \mathbb{R}_+$. Also $I_q(\cdot)$ is the Bessel function of order q :

$$I_q(x) = \sum_{k=0}^{\infty} \left(\frac{x}{2}\right)^{2k+q} \frac{1}{k! \Gamma(k+q+1)}, \quad x \in \mathbb{R}$$

Since we have conditional density function, we can estimate parameters numerically.

3.2 Approximated likelihood methods

Another method of deriving estimators involves employing approximation techniques. Rather than directly approximating the transition density, these techniques approximate the trajectory of the process. This is done so that the discretized version of the process yields a usable likelihood. We can take all numerical schemes, developed in Section 2 and apply them to the CIR process and the Ait-Sahalia model directly or using transformation with Ito formula, if needed.

Firstly let us take CIR process of the form (3):

- **Truncated scheme** If we simulate the solution to the CIR process with the truncated scheme of the form (18), distribution of the transition of the discrete process $\hat{X}_n$ is given by:

$$\hat{X}_{n+1} | \hat{X}_n = x \sim \mathcal{N}\left(x + \Delta(a - k \cdot \min(x, \frac{3}{\Delta^{\frac{1}{6}}} - 1)), \sigma^2 x \Delta\right).$$

Let us denote $c := \frac{3}{\Delta^{\frac{1}{6}}} - 1$ and write down the transition density of the process $\hat{X}_n$:

$$\rho_{\hat{X}_{n+1}}(y|\theta, \hat{X}_n = x) = \frac{1}{\sqrt{2\pi\sigma^2 x \Delta}} \exp\left(-\frac{1}{2} \cdot \frac{(y - x - \Delta(a - k \cdot \min(x, c)))^2}{\sigma^2 x \Delta}\right) \quad (42)$$

Hence the log-likelihood will have the form:

$$l_N(\theta) = \sum_{i=0}^{N-1} -\frac{1}{2} \frac{(\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c)))^2}{\sigma^2 \hat{X}_i \Delta} - \sum_{i=0}^{N-1} \frac{1}{2} \ln(2\pi\sigma^2 \hat{X}_i \Delta) \quad (43)$$

First order condition:

$$\begin{cases} \frac{\partial l_N(\theta)}{\partial a} = \sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c))}{\sigma^2 \hat{X}_i \Delta} \cdot \Delta = 0 \\ -\frac{\partial l_N(\theta)}{\partial k} = -\sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c))}{\sigma^2 \hat{X}_i \Delta} \cdot \Delta \min(\hat{X}_i, c) = 0 \\ \frac{\partial l_N(\theta)}{\partial \sigma^2} = \sum_{i=0}^{N-1} \frac{1}{2} \frac{(\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c)))^2}{\sigma^4 \hat{X}_i \Delta} - \sum_{i=0}^{N-1} \frac{1}{2\sigma^2} = 0 \end{cases} \quad (44)$$

We can rewrite it as

$$\begin{cases} \sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c))}{\hat{X}_i} = 0 \\ \sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c))}{\hat{X}_i} \cdot \min(\hat{X}_i, c) = 0 \\ \sigma^2 = \frac{N-1}{N} \sum_{i=0}^{N-1} \frac{(\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c)))^2}{\hat{X}_i} \end{cases} \quad (45)$$

Taking first and second equations, let us derive estimators for a and k :

$$\begin{cases} \sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i}{\hat{X}_i} = a \Delta \sum_{i=0}^{N-1} \frac{1}{\hat{X}_i} - k \Delta \sum_{i=0}^{N-1} \frac{\min(\hat{X}_i, c)}{\hat{X}_i} \\ \sum_{i=0}^{N-1} \frac{(\hat{X}_{i+1} - \hat{X}_i) \cdot \min(\hat{X}_i, c)}{\hat{X}_i} = a \Delta \sum_{i=0}^{N-1} \frac{\min(\hat{X}_i, c)}{\hat{X}_i} - k \Delta \sum_{i=0}^{N-1} \frac{\min^2(\hat{X}_i, c)}{\hat{X}_i} \end{cases} \quad (46)$$

$$\begin{cases} \sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i}{\hat{X}_i} = a \Delta \sum_{i=0}^{N-1} \frac{1}{\hat{X}_i} - k \Delta \sum_{i=0}^{N-1} \frac{\min(\hat{X}_i, c)}{\hat{X}_i} \\ k \Delta = \frac{1}{\sum_{i=0}^{N-1} \frac{\min^2(\hat{X}_i, c)}{\hat{X}_i}} \left(a \Delta \sum_{i=0}^{N-1} \frac{\min(\hat{X}_i, c)}{\hat{X}_i} - \sum_{i=0}^{N-1} \frac{(\hat{X}_{i+1} - \hat{X}_i) \cdot \min(\hat{X}_i, c)}{\hat{X}_i} \right) \end{cases} \quad (47)$$

Hence we can derive

$$\hat{a} = \frac{1}{\Delta} \left(\frac{\sum_{i=1}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i}{\hat{X}_i} - \frac{(\sum_{i=1}^{N-1} \frac{\min(\hat{X}_i, c)}{\hat{X}_i})(\sum_{i=1}^{N-1} \frac{(\hat{X}_{i+1} - \hat{X}_i) \cdot \min(\hat{X}_i, c)}{\hat{X}_i})}{\sum_{i=1}^{N-1} \frac{\min^2(\hat{X}_i, c)}{\hat{X}_i}}}{\sum_{i=1}^{N-1} \frac{1}{\hat{X}_i} - \frac{(\sum_{i=1}^{N-1} \frac{\min(\hat{X}_i, c)}{\hat{X}_i})^2}{\sum_{i=1}^{N-1} \frac{\min^2(\hat{X}_i, c)}{\hat{X}_i}}} \right). \quad (48)$$

$\hat{k}$ can be derived after substituting $\hat{a}$ into (47), $\hat{\sigma}$ - after substituting $\hat{a}$ and $\hat{k}$ into (45). Now we can estimate accuracy of derived estimators. We take $a = 60, k = 20, \sigma = 1$ and simulate the approximation to the true solution of the CIR process with given parameters using truncated scheme. Now we can check estimators coincide with its theoretical values.

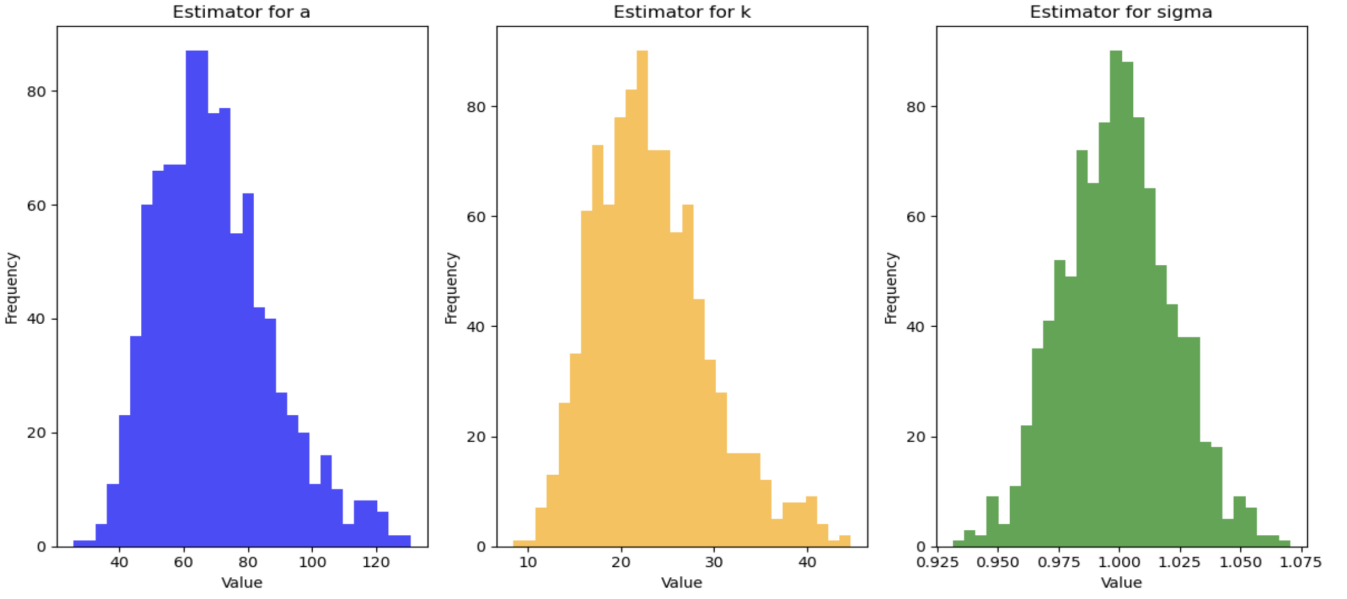


Figure 7: Distribution of estimators for mean and sigma after 1000 simulations. Source: author's calculations

• Tamed scheme

Previously we proved that tamed scheme can not be directly applied to the CIR process, that is why we need to transform it first. Take $Z_t = (X_t)^{\frac{1}{2}}$, where X_t is the CIR process (3). Applying Ito formula (Definition 1.7), we get:

$$dZ_t = \frac{1}{2(X_t)^{\frac{1}{2}}}dX_t - \frac{1}{8}(X_t)^{-\frac{3}{2}}\langle X \rangle_t = \frac{1}{2}(X_t)^{-\frac{1}{2}}(a - kX_t - \frac{\sigma^2}{4})dt + \frac{\sigma}{2}dW_t =$$

$$\frac{1}{2}\left(\frac{a - \frac{\sigma^2}{4}}{Z_t} - kZ_t\right)dt + \frac{\sigma}{2}dW_t \quad (49)$$

Statement 3.1. Function $f(x) = \frac{1}{x}$ is one-sided Lipschitz continuous (Definition 1.3).

Proof. $(x - y)(f(x) - f(y)) = \frac{-(x-y)^2}{xy}$. Since $x, y \geq 0$, $(x - y)(f(x) - f(y)) \leq 0$, hence for any $L > 0$

$$(x - y)(f(x) - f(y)) \leq L(x - y)^2$$

□

Drift function is one-sided Lipschitz since it is a sum of linear and hyperbolic functions. Diffusion function is constant, hence all the assumptions of the tamed scheme are satisfied and we can apply it to Z_t :

$$\hat{Z}_{n+1} = \hat{Z}_n + \frac{\Delta \cdot f(\hat{Z}_n)}{1 + \Delta \cdot |f(\hat{Z}_n)|} + \frac{\sigma}{2}(W_{t_{n+1}} - W_{t_n}), \hat{Z}_0 = x_0^{\frac{1}{2}} \quad (50)$$

where $f(x) = \frac{1}{2}\left(\frac{a - \frac{\sigma^2}{4}}{x} - kx\right)$. We can write down conditional law for the process $\hat{Z}_n$:

$$\hat{Z}_{n+1}|\hat{Z}_n = z \sim \mathcal{N}\left(z + \frac{\Delta \cdot f(z)}{1 + \Delta \cdot |f(z)|}, \frac{\Delta \sigma^2}{4}\right) \quad (51)$$

Since the diffusion function does not depend on z , we can write down an estimator for σ :

$$\hat{\sigma} = 2 \cdot \sqrt{\sum_{i=0}^{N-1} (\hat{Z}_{i+1} - \hat{Z}_i)^2}. \quad (52)$$

Now we can write down the transition law of the process $\hat{Z}_n$ and the log-likelihood:

$$\rho_{\hat{Z}_{n+1}}(y|\theta, \hat{Z}_n = z) = \frac{1}{\sqrt{\pi \sigma^2 \Delta / 2}} \exp\left(-\frac{1}{2} \cdot \frac{(y - z - \frac{\Delta \cdot f(z)}{1 + \Delta \cdot |f(z)|})^2}{\frac{\Delta \sigma^2}{4}}\right) \quad (53)$$

$$l_N(\theta) = \sum_{i=0}^{N-1} -\frac{1}{2} \frac{(\hat{Z}_{i+1} - \hat{Z}_i - \frac{\Delta \cdot f(\hat{Z}_i)}{1 + \Delta \cdot |f(\hat{Z}_i)|})^2}{\frac{\Delta \sigma^2}{4}} - \sum_{i=0}^{N-1} \frac{1}{2} \ln(\pi \sigma^2 \Delta / 2). \quad (54)$$

Since $\hat{\sigma}$ is estimated, the maximization of the log-likelihood (54) is equivalent to the maximization of

$$\sum_{i=0}^{N-1} (\hat{Z}_{i+1} - \hat{Z}_i) \cdot \frac{\Delta f(\hat{Z}_i)}{1 + \Delta |\hat{Z}_i|} - \frac{1}{2} \cdot \left(\frac{\Delta f(\hat{Z}_i)}{1 + \Delta |\hat{Z}_i|} \right)^2. \quad (55)$$

This optimization can be done numerically with Python 3.9 and `scipy.minimize` function with method L-BFGS-B, where "L-BFGS" stands for "Limited-memory Broyden–Fletcher–Goldfarb–Shanno" and "B" denotes that this variant allows for bounds on the variables.

Given a function $f(\mathbf{x})$ that we want to minimize, the algorithm proceeds by approximating the inverse Hessian matrix of second derivatives via updates specified by gradient evaluations. The main steps of the L-BFGS-B algorithm are as follows (e.g. Byrd et al, 1994):

1. We start with an initial guess x_0 and initialize parameters including the history size m , which determines how many previous gradient and position vectors are stored.
2. After the gradient $\nabla f(x_k)$ is computed, the inverse Hessian matrix should be approximated. L-BFGS updates is used, relying on the most recent m updates to approximate the inverse Hessian. Algorithm uses the approximated inverse Hessian to determine the search direction:

$$p_k = -H_k \nabla f(x_k)$$

3. Perform a line search to find an optimal step size α_k that satisfies certain conditions (e.g., Wolfe conditions), then update:

$$x_{k+1} = x_k + \alpha_k p_k$$

4. L-BFGS-B specifically addresses variable bounds. The algorithm modifies the search directions and step sizes to ensure the updated variables remain within specified bounds.

The algorithm may terminate based on several criteria, often either when the maximum number of iterations is exceeded, the change in function value is beneath a specified tolerance, or when the gradient norm falls below a specified tolerance, indicating convergence.

As for the truncated scheme, we can empirically test accuracy of the estimators.

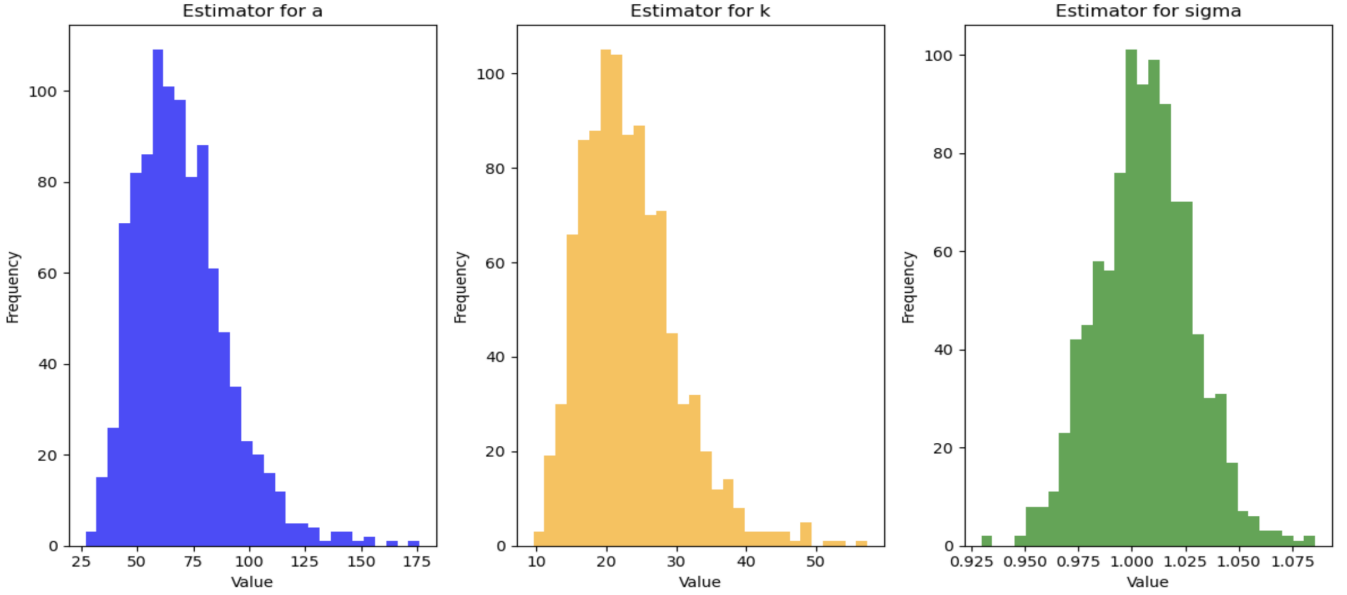


Figure 8: Distribution of estimators for mean and sigma after 1000 simulations. Source: author's calculations

Exponential scheme can be directly applied to the Ait-Sahalia model (6) with condition $\beta > 1$. The approximation of the true solution is

$$\hat{X}_{n+1} = \hat{X}_n \cdot \exp\left(\left(b - \frac{\sigma^2}{2} \hat{X}_n^{2(\beta-1)}\right)\Delta + \sigma \hat{X}_n^{\beta-1} (W_{t_{n+1}} - W_{t_n})\right). \quad (56)$$

Since $\hat{X}_n$ stay positive for every n , we can take logarithm of both parts of the equation and write down the law of the transition of the scheme:

$$\ln \hat{X}_{n+1} | \hat{X}_n = x \sim \mathcal{N}\left(\ln x + \Delta\left(b - \frac{\sigma^2}{2} x^{2(\beta-1)}\right), \sigma^2 x^{2(\beta-1)} \Delta\right).$$

The transition density of the process is

$$\rho_{\ln \hat{X}_{n+1}}(y | \theta, \hat{X}_n = x) = \frac{1}{\sqrt{2\pi\sigma^2 x^{2(\beta-1)} \Delta}} \exp\left(-\frac{1}{2} \cdot \frac{(y - \ln x - (b - \frac{\sigma^2}{2} x^{2(\beta-1)})\Delta)^2}{\sigma^2 x^{2(\beta-1)} \Delta}\right). \quad (57)$$

The log-likelihood of the transition is

$$l_N(\theta) = \sum_{i=0}^{N-1} \left(-\ln(\sigma^2 \hat{X}_i^{2(\beta-1)} \Delta) - \frac{1}{2} \cdot \frac{(\ln \hat{X}_{i+1} - \ln \hat{X}_i - (b - \frac{\sigma^2}{2} \hat{X}_i^{2(\beta-1)})\Delta)^2}{\sigma^2 \hat{X}_i^{2(\beta-1)} \Delta} \right). \quad (58)$$

Now we can maximize log-likelihood numerically. As for the previous schemes, let us test the accuracy of the estimators on simulated paths of the process. Take $b = \sigma = 1, \beta = \frac{3}{2}$. Results of the simulation are presented in the Figure 9.

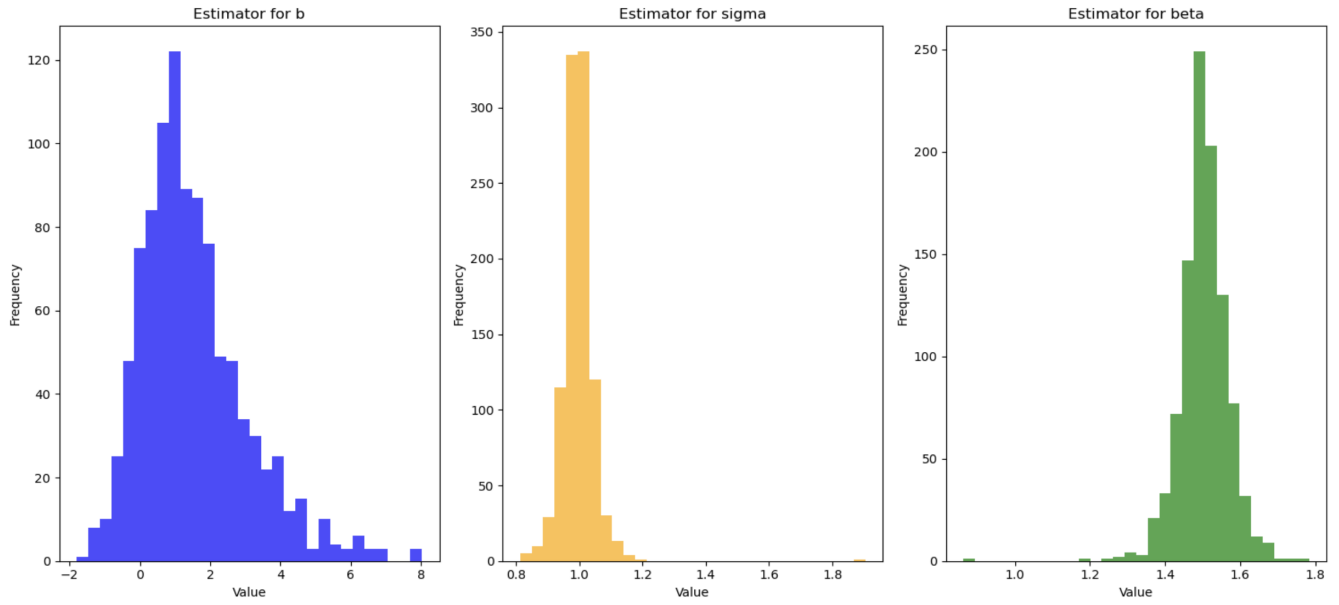


Figure 9: Distribution of estimators for b , σ and β after 1000 simulations. Source: author's calculations

4 Application to Libor and SOFR

This section is dedicated to Libor and SOFR interest rates. Part 1 includes history and financial importance of the Libor market, Part 2 is about applying all the methods defined in previous chapters to the real data.

4.1 History of the Libor market

The London Interbank Offered Rate, or Libor, has been a fundamental benchmark in financial markets around the world since the 1980s. Originating in London, this rate has guided the pricing of various financial instruments including mortgages and derivatives. Introduced in 1986 by the British Bankers' Association, Libor was created to standardize the pricing of loans and financial products across international banks. It began with the US dollar and extended to other currencies, representing the rate banks charge each other for short-term loans in the London interbank market.

Libor influences trillions of dollars' worth of financial products globally. It serves as a reference point for setting interest rates on products ranging from personal loans to complex financial derivatives. The reliance on Libor highlights its key role in the finance sector, where it acts as a measure of bank health and helps set rates for various consumer finance products.

The reliability of Libor was questioned following the financial crisis of 2007-2008, leading to a scandal in 2012 that exposed manipulation by some banks. These issues prompted calls for a reform to replace Libor with more transparent and reliable alternatives. New benchmarks like the Secured Overnight Financing Rate (SOFR) in the US and the Sterling Overnight Index Average (SONIA) in the UK were introduced.

Both Libor and SOFR serve as reference rates that influence vast amounts of financial products, but they differ in their calculation, underlying assets, and risk characteristics. The key distinctions between Libor and SOFR are:

1. Libor is based on the estimated interest rates at which banks could borrow unsecured funds from other banks in the London interbank market. Libor rates are determined through submissions by a panel of banks, estimating the rates at which they would be able to borrow funds for different time periods (overnight to 12 months) across several currencies. In this work we are considering USD as a currency and overnight as a period. In contrast, SOFR is derived from actual transactions in the U.S. Treasury repurchase agreement (repo) market, where banks and investors borrow or lend Treasuries overnight. SOFR thus reflects the cost of borrowing cash overnight collateralized by U.S. Treasury securities.
2. Being an unsecured rate, Libor carries a credit risk component, reflecting the risk that the lending bank might not be repaid. This risk factor is inherently built into the rate. As a secured rate, backed by U.S. Treasury securities, SOFR contains virtually no credit risk. This makes it a safer, more stable rate from a risk perspective.
3. The reliability of Libor has been questioned due to it being susceptible to manipulation. Since Libor is based on estimates rather than actual transactions, the rate can be influenced by the submitting banks' perceptions and integrity. SOFR is considered more reliable and reflective of the market since it is based on observable transactions in a highly liquid, deep market. There

are billions of dollars in daily transactions supporting the SOFR rate, providing a robust, tamper-resistant foundation.

4. Historically, Libor has been embedded in a wide range of financial products including mortgages, loans, and derivatives. Its widespread use across different currencies and terms provides flexibility in financial contracts. Currently, SOFR is primarily used in the U.S. dollar markets and covers only overnight transactions. Its adoption is increasing, but transitioning from Libor involves restructuring existing contracts and adapting to using a purely overnight rate, which can be challenging for products previously based on term rates like Libor.
5. It is a forward-looking rate, quoted at the beginning of the borrowing period. It provides predictability in terms of interest costs. SOFR is inherently a backward-looking rate, calculated at the end of the borrowing period based on completed transactions. This could potentially lead to uncertainty in interest obligations, though efforts are underway to develop a forward-looking SOFR term rate.

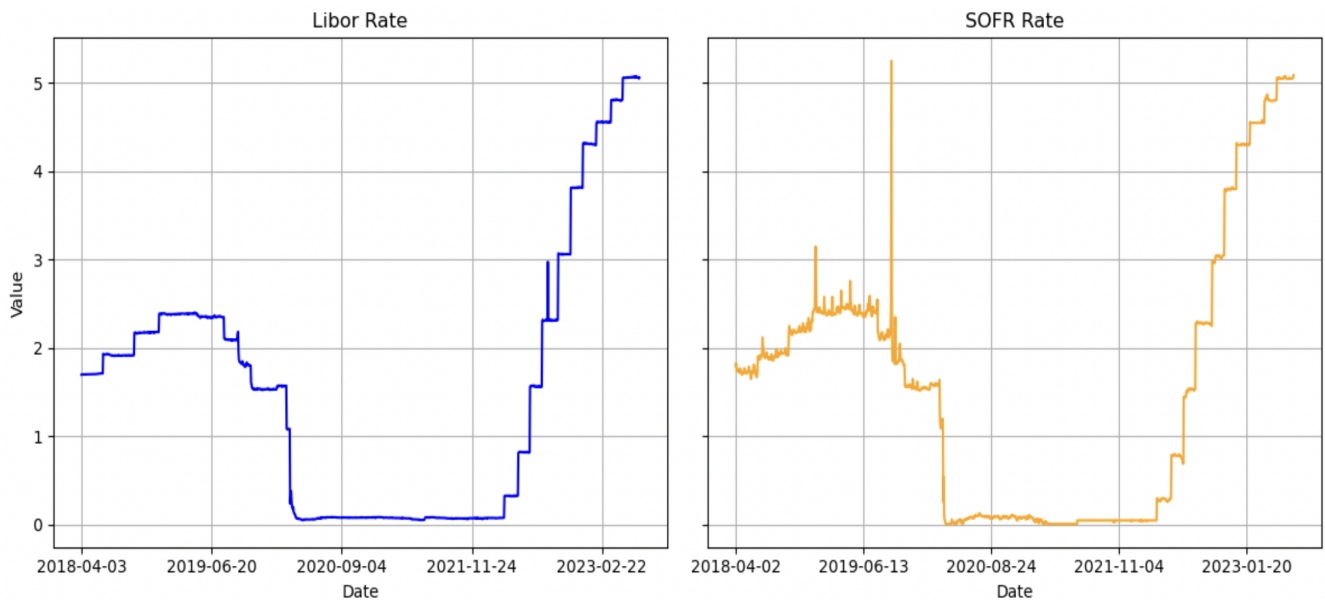


Figure 10: Libor and SOFR rates. Source: iborate.com, newyorkfed.org

The graph presents both the Libor and SOFR rates over the time period from April 3, 2018, the inception date of SOFR, to June 30, 2023, the final day of Libor's existence. It can be seen that SOFR rate is more volatile and has more outliers, but general dynamics of two rates coincides. Hence we can concatenate data on Libor and SOFR and use models, fitted on the historical Libor data, to predict SOFR dynamic.

4.2 Parameter estimation on the Libor data

Since we have derived estimators for the CIR process and the Ait-Sahalia model parameters, we can test it on the daily USD Libor data from 02.01.2001 to 01.03.2022. Results of the estimation are presented in Table 2 and Table 3.

Method	$\hat{a}$	$\hat{k}$	$\hat{\sigma}$
(1) Exact likelihood	17.555	15.824	5.418
(2) Truncated scheme	2.29	5.776	5.566
(3) Tamed scheme	9.897	10.733	5.411

Table 2: Parameter estimation results for the CIR process. Source: author's calculations

Method	RMSE	MAE	MAPE
(1) Exact likelihood, CIR	1.16	0.95	1.93
(4) Exponential scheme, Ait-Sahalia	1	0.89	0.49

Table 3: Parameter estimation results for the CIR process. Source: author's calculations

It can be seen from the tables that all estimators for the CIR process have theoretical mean between 0.3 and 1.2 and $\hat{\sigma}$ around 4.8 - 5.1. Only for exact likelihood estimation of parameters condition $2a > \sigma^2$ is satisfied. Hence if we will try to model the approximation to the true solution of the CIR process with parameters (2) or (3), there is no guarantee that approximation stays positive, so we will not use these parameters.

Method	$\hat{b}$	$\hat{\sigma}$	$\hat{\beta}$
(4) Exponential scheme	4.479	4.093	1.133

Table 4: Parameter estimation results for the Ait-Sahalia model. Source: author's calculations

In contrast, all conditions for the Ait-Sahalia model with parameters (4) are satisfied. Since $\hat{b} = 4 > 0$, drift function is always positive, hence the process have a tendency to grow.

4.3 Interest rate swaps pricing

Interest rate swaps are one of the most prevalent financial derivatives used in the market. They involve the exchange of interest rate cash flows between two parties based on a specified principal amount, which, however, does not exchange hands. These instruments help both financial and non-financial entities manage their exposure to interest rate volatility, transform their liability structures, or attain a comparative advantage in borrowing.

There are primarily two types of interest rate swaps:

1. Fixed-for-Fixed Swaps: Both parties exchange interest payments at fixed rates, possibly in different currencies.
2. Fixed-for-Floating Swaps: One party pays a fixed rate, while the other pays a floating rate, typically referenced to an index like the LIBOR, EURIBOR, or the federal funds rate.

Further in this work we will focus on the fixed-for-floating 1 year single currency swaps. In a typical fixed-for-floating single currency swap, the principal amount upon which the swap is based remains constant over the life of the swap and serves purely as a reference point for calculating the periodic

interest obligations. The fixed payer agrees to pay a steady interest rate on this principal, which remains unchanged through the course of the swap. On the other side, the floating payer's rate resets periodically, based on a predetermined benchmark, reflecting current market interest rates.

We will consider the swap pricing model based on swap's present value estimation (e.g. Hull, 2012), where present value of floating payment receiver (PV_r) is equal to the present value of the floating payment payer (PV_p) due to no arbitrage opportunity. Present values of both sides can be derived as

$$\begin{cases} PV_r = S \cdot \sum_{i=1}^{12} \frac{P \cdot df_i}{12} \\ PV_p = S \cdot \sum_{i=1}^{12} \frac{r_i \cdot df_i}{12}, \end{cases} \quad (59)$$

where S is a sum of the contract, the base of payments calculating for both fixed and floating rates; P is the price of the swap at the time t_0 , in our case it is 01.03.2022; r_i is SOFR rate at the time t_i ; df_i is discount factor at the period i that can be calculated as

$$\begin{cases} df_{i+1} = \frac{df_i}{1 + \frac{r_i}{12}} \\ df_0 = 1. \end{cases} \quad (60)$$

Since we consider 1 year swap with monthly payments, indices in (62) and (63) represent number of month starting from the April, 2022. Hence by the SOFR rate at the time t_i we mean value of the SOFR rate at the first business day of month i . Now we can calculate the cashflow of the swap for the payer side:

$$CF_p = PV_p - PV_r = \frac{S}{12} \sum_{i=1}^{12} (r_i - P) \cdot df_i, \quad (61)$$

$CF_r = -CF_p$ by no-arbitrage condition. Now we can all the frameworks for SOFR forecasting in order to calculate $\mathbb{E}[CF_p]$, since r_i is the only unknown variable, specifically CIR process with parameters (1) and Ait-Sahalia model with parameters (4).

5 Conclusion

All the research's task were achieved: for every possible combination of an SDE and numerical scheme for the true solution approximation we defined estimators for model's parameters. Since the CIR process and the Ait-Sahalia model have nontrivial form, basic Euler-Maruyama scheme could not be used, so we considered advanced numerical schemes, such as tamed, truncated, exponential schemes and implicit scheme by Alfonsi. In cases where we were not be able to directly apply a numerical scheme to the process, we used transformation of a process with Ito formula, for example for combination of the CIR process and the tamed scheme.

All the parameter estimations with approximation of the true solution insted of the exact solution were made via log-likelihood maximixation. In case of combination of the CIR process and the truncated scheme we derived the estimators analytically. In other cases we did optimisation with algorithms on Python. In all combinations of model and numerical scheme we were able to empirically check if derived estimators are accurate. Since they are centered symmetrically around the true values and have small tales, we can state that these estimators are accurate.

When applying estimated models to the true Libhor data, we could have compared them. Ait-Sahalia model seems more preferable in approximating Libor interest rate, since it pas power ≥ 1 in diffusion function and it's drift is always positive, hence this model can faster adapt to changes of a short term interest rate quickly.

References

- [1] Aurélien Alfonsi (2005): On the discretization schemes for the CIR (and Bessel squared) processes. Monte Carlo Methods and Applications.
- [2] Desmond J. Higham (2001): An Algorithmic Introduction to Numerical Simulation of Stochastic Differential Equations. SIAM Review 43(3):525-546
- [3] Hao Yang, Fuke Wu, Peter E. Kloeden, Xuerong Mao (2020): The truncated Euler–Maruyama method for stochastic differential equations with Hölder diffusion coefficients. Journal of Computational and Applied Mathematics 366:112379
- [4] John C. Cox, Jonathan E. Ingersoll, Stephen A. Ross (1985): A Theory of the Term Structure of Interest Rates. Econometrica 53(2):385-407
- [5] John C. Hull (2012): Options, Futures, and Other Derivatives. Prentice Hall
- [6] Richard H. Byrd, Peihuang Lu, Jorge Nocedal, Ciyu Zhu (1994): A Limited Memory Algorithm for Bound Constrained Optimization. SIAM Journal on Scientific Computing 16(5)
- [7] Stefano M. Iacus (2008): Simulation and Inference for Stochastic Differential Equations. Springer Series in Statistics
- [8] Martin Hutzenthaler, Aenulf Jentzen, Peter E. Kloeden (2012): Strong convergence of an explicit numerical method for SDES with nonglobally Lipschitz continuous coefficients. The Annals of Applied Probability 22(4)
- [9] Mireille Bossy, Jean-François Jabir, Kerlyns Martinez (2021): On the weak convergence rate of an exponential Euler scheme for SDEs governed by coefficients with superlinear growth. Bernoulli 27(1):312-347
- [10] Oldřich Vašíček (1977): An equilibrium characterization of the term structure. Finance, Economics and Mathematics (pp.33-44)
- [11] Sotirios Sabanis (2016): Euler approximations with varying coefficients: the case of superlinearly growing diffusion coefficient. The Annals of Applied Probability 26(4)
- [12] Xuerong Mao (2015): The truncated Euler–Maruyama method for stochastic differential equations. Journal of Computational and Applied Mathematics 290(3)
- [13] Yacine Ait-Sahalia (1996): Testing Continuous-Time Models of the Spot Interest Rate. The Review of Financial Studies Vol. 9, No. 2 (Summer, 1996), pp. 385-426
- [14] iborate.com
- [15] newyorkfed.org

Appendix A: Python code for the CIR process parameter estimation

```
1 class CIR:
2     def __init__(self, method: str, parameters: tuple, x_0: float, N: int, T=1,
3     true_X=[]):
4         self.method = method
5         self.true_parameters = parameters
6         self.estimated_parameters = ()
7         self.x_0 = x_0
8         self.N = N
9         self.T = T
10        self.dt = T / N
11        self.X = np.zeros(self.N)
12        self.true_X = true_X
13
14        self.sigma_hat = 1
15
16    def make_simulation(self):
17        a, k, sigma = self.true_parameters
18        dW = np.random.normal(0, np.sqrt(self.dt), size=self.N)
19
20        self.X = np.zeros(self.N)
21        self.X[0] = self.x_0
22
23        if self.method == 'truncated':
24            f_cir = lambda x: a - k * x
25            c = 3 / (self.dt ** (1 / 6)) - 1
26            f = lambda x: f_cir(min(x, c))
27            g = lambda x: sigma * np.sqrt(x)
28
29        elif self.method == 'tamed':
30            f = lambda x: (((a - (sigma ** 2) / 4) / x) - k * x) / 2
31            g = lambda x: sigma / 2
32
33        for j in range(1, self.N):
34            self.X[j] = self.X[j-1] + f(self.X[j-1]) * self.dt + g(self.X[j - 1])
35            * dW[j - 1]
36
37        return
38
39    def expBes(self, nu, x):
40        mu = 4 * nu ** 2
41        A1 = 1
42        A2 = A1 * (mu - 1) / (1 * (8 * x))
43        A3 = A2 * (mu - 9) / (2 * (8 * x))
44        A4 = A3 * (mu - 25) / (3 * (8 * x))
45        A5 = A4 * (mu - 49) / (4 * (8 * x))
46        A6 = A5 * (mu - 81) / (5 * (8 * x))
47        A7 = A6 * (mu - 121) / (6 * (8 * x))
48
49        return 1 / np.sqrt(2 * np.pi * x) * (A1 - A2 + A3 - A4 + A5 - A6 + A7)
50
51    def cdfCIR(self, x, t, x0, theta):
52        c = 2 * theta[1] / ((1 - np.exp(-theta[1] * t)) * theta[2] ** 2)
```

```

53     ncp = 2 * c * x0 * np.exp(-theta[1] * t)
54     df = 4 * theta[0] / theta[2] ** 2
55     u = c * x0 * np.exp(-theta[1] * t)
56     v = c * x
57     q = 2 * theta[0] / theta[2] ** 2 - 1
58
59     lik = (np.log(c) - (u + v) + (q / 2) * np.log(v / u) +
60           np.log(self.expBes(q, 2 * np.sqrt(u * v))) + 2 * np.sqrt(u * v))
61
62     return lik
63
64 def GaussianApproximation(self, theta):
65     a_hat, k_hat = theta
66     f = lambda x: (((a_hat - (self.sigma_hat ** 2) / 4) / x) - k_hat * x) / 2
67     drift = lambda x: (f(x) / self.N) / (1 + abs(f(x)) / self.N)
68
69     return -sum([(self.X[i + 1] - self.X[i]) * drift(self.X[i]) -
70                  ((drift(self.X[i])) ** 2) / 2
71                  for i in range(len(self.X) - 1)])
72
73 def estimate_parameters(self):
74
75     if sum(self.X) == 0:
76         self.X = self.true_X
77
78     if self.method == 'truncated':
79
80         m = lambda x: np.where(x < 3 / (self.dt ** (1 / 6)) - 1, x, 3 / (self
81 .dt ** (1 / 6)) - 1)
82
83         a_hat = (self.N * ((np.sum((self.X[1:] - self.X[:-1]) / self.X[:-1])
84 - np.sum(m(self.X[:-1])
85           / self.X[:-1]) * np.sum((self.X[1:] - self.X[:-1]) * m(self.X
86[:-1]) /
87           self.X[:-1]) / np.sum(np.square(m(self.X[:-1])) / self.X
88[:-1])))
89           / (np.sum(1 / self.X[:-1]) - ((np.sum(m(self.X[:-1])
90 / self.X[:-1])) ** 2) / np.sum(np.square(m(self.X[:-1])) /
91 self.X[:-1]))))
92
93         k_hat = (((a_hat * self.dt * np.sum(m(self.X[:-1]) / self.X[:-1]) -
94 np.sum((self.X[1:] - self.X[:-1]) * m(self.X[:-1]) /
95 self.X[:-1])) / (np.sum(np.square(m(self.X[:-1])) /
96 self.X[:-1])))) / self.dt)
97
98         sigma_hat = (np.sqrt(np.sum(np.square((self.X[1:] - self.X[:-1] -
99 self.dt *
100           (a_hat - k_hat * m(self.X[:-1])))) / (self.X[:-1])) / (
101 self.dt * (self.N - 1))))
102
103         self.estimated_parameters = a_hat, k_hat, sigma_hat
104
105     elif self.method == 'exact_likelihood':
106
107         likelihood = lambda params: -np.sum(self.cdfCIR(self.true_X[1:],
108           self.dt * np.ones(len(self.true_X
109) - 1),
110           self.true_X[:-1],

```

```

104         params))
105     initial_params = np.random.uniform(1, 10, 3)
106     result = minimize(likelihood, initial_params, method='L-BFGS-B')
107     self.estimated_parameters = result.x
108
109     elif self.method == 'tamed':
110
111         initial_params = np.random.uniform(1, 10, 2)
112         sigma_hat = np.sqrt(np.sum(np.square(self.X[1:] - self.X[:-1]))) * 2
113         self.sigma_hat = sigma_hat
114         result = minimize(self.GaussianApproximation, initial_params, method=
'L-BFGS-B')
115         self.estimated_parameters = (result.x[0], result.x[1], sigma_hat)
116
117     return

```

Listing 1: CIR process parameter estimation

Appendix B: Python code for Ait-Sahalia model parameter estimation

```
1
2 class AitSahalia:
3     def __init__(self, parameters: tuple, x_0: float, N: int, T=1):
4         self.true_parameters = parameters
5         self.estimated_parameters = ()
6         self.x_0 = x_0
7         self.N = N
8         self.T = T
9         self.dt = T / N
10        self.X = np.zeros(self.N)
11
12    def make_simulation(self):
13        b, sigma, beta = self.true_parameters
14        f = lambda x: b * x
15        dW = np.random.normal(0, np.sqrt(self.dt), size=self.N)
16        self.X = np.zeros(self.N)
17        self.X[0] = self.x_0
18
19        for j in range(1, self.N):
20            self.X[j] = (self.X[j-1] * np.exp(sigma * (self.X[j-1]) ** (beta - 1)
21 * dW[j - 1] +
22            (b - (sigma ** 2) * (self.X[j - 1] ** (2 * (beta - 1))) / 2) * self.
23 dt))
24
25        return
26
27    def cdfEXP(self, y, t, x, theta):
28
29        mean = (theta[0] - (theta[1] ** 2) * (x ** (2 * (theta[2] - 1))) / 2) * t
30        std = theta[1] * (x ** (theta[2] - 1)) * np.sqrt(t)
31
32        y = np.log(y) - np.log(x)
33
34        lik = (((y - mean) / std) ** 2) / 2 + np.log(np.sqrt(2 * np.pi * (std
35 ** 2)))
36
37        return lik
38
39    def estimate_parameters(self, X=None):
40
41        if X is not None:
42            self.X = X
43
44        likelihood = lambda params: np.sum(self.cdfEXP(self.X[1:],
45            self.dt * np.ones(len(self.X) - 1),
46            self.X[:-1], params))
47
48        initial_params = np.random.uniform(1, 10, 3)
49        result = minimize(likelihood, initial_params, method='L-BFGS-B')
50        self.estimated_parameters = (result.x[0], abs(result.x[1]), result.x[2])
51
52        return
```

Listing 2: Python code for Ait-Sahalia model parameter estimation

Appendix C: Hessian matrix calculation

In the maximum likelihood estimation of the parameters in a framework of the CIR process and the tamed scheme, we have a task:

Hence the log-likelihood will have the form:

$$l_N(\theta) = \sum_{i=0}^{N-1} -\frac{1}{2} \frac{(\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c)))^2}{\sigma^2 \hat{X}_i \Delta} - \sum_{i=0}^{N-1} \frac{1}{2} \ln(2\pi\sigma^2 \hat{X}_i \Delta),$$

where first order condition are:

$$\begin{cases} \frac{\partial l_N(\theta)}{\partial a} = \sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c))}{\sigma^2 \hat{X}_i \Delta} \cdot \Delta = 0 \\ -\frac{\partial l_N(\theta)}{\partial k} = -\sum_{i=0}^{N-1} \frac{\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c))}{\sigma^2 \hat{X}_i \Delta} \cdot \Delta \min(\hat{X}_i, c) = 0 \\ \frac{\partial l_N(\theta)}{\partial \sigma^2} = \sum_{i=0}^{N-1} \frac{1}{2} \frac{(\hat{X}_{i+1} - \hat{X}_i - \Delta(a - k \cdot \min(\hat{X}_i, c)))^2}{\sigma^4 \hat{X}_i \Delta} - \sum_{i=0}^{N-1} \frac{1}{2\sigma^2} = 0 \end{cases}$$

After we derived the estimators, we should check if Hessian matrix

$$H = \begin{pmatrix} \frac{\partial^2 l_N}{\partial a^2} & \frac{\partial^2 l_N}{\partial a \partial k} & \frac{\partial^2 l_N}{\partial a \partial \sigma^2} \\ \frac{\partial^2 l_N}{\partial a \partial k} & \frac{\partial^2 l_N}{\partial k^2} & \frac{\partial^2 l_N}{\partial k \partial \sigma^2} \\ \frac{\partial^2 l_N}{\partial a \partial \sigma^2} & \frac{\partial^2 l_N}{\partial k \partial \sigma^2} & \frac{\partial^2 l_N}{\partial (\sigma^2)^2} \end{pmatrix}$$

is negatively definite. By the Sylvester's criterion matrix is negative semi-definite if $\Delta_1 < 0$, $\Delta_2 > 0$ and $\Delta_3 < 0$.

$$\Delta_1 = -\sum_{i=1}^{N-1} \frac{\Delta}{\sigma^2 \hat{X}_i} < 0$$

$$\Delta_2 = \left(\sum_{i=1}^{N-1} \frac{\Delta}{\sigma^2 \hat{X}_i} \right)^2 - \left(\sum_{i=1}^{N-1} \frac{\Delta \cdot \min(\hat{X}_i, c)}{\sigma^2 \hat{X}_i} \right)^2$$

Sign of Δ_2 depends on c , and in our specific case Δ_2 is negative. So that is why we should check if log-likelihood of initial parameters was the biggest during the optimization.